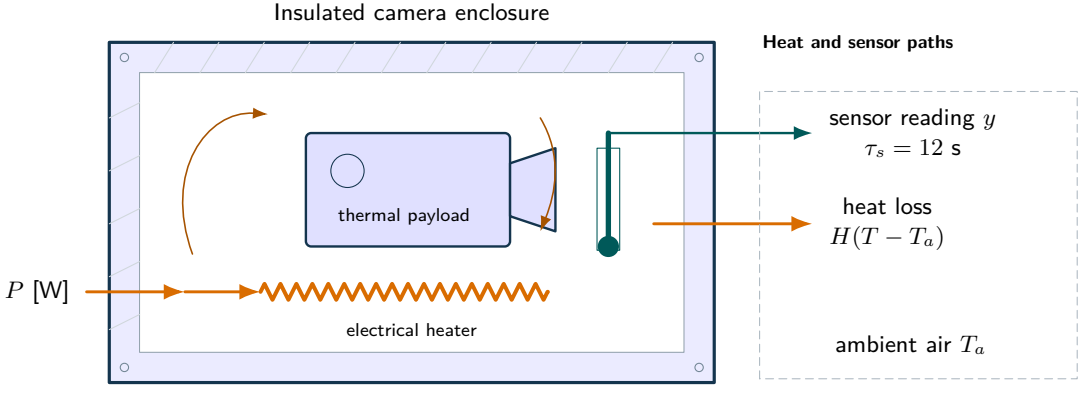

Stored and indicated heat

$x = T - T_a$ [K]; y [K].
Physical storage and sensor memory are separate states.

Balances

$900\dot{x} = P - 15x$.
$12\dot{y} = x - y$.
Heat rate in W; time in s.

Read the cutaway

Orange: heat flow. Teal: measurement. Wall and payload are one lumped capacity; no spatial field is resolved.